

Smart Inventory Management System

AI-Based Inventory Movement & Stock Monitoring System

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Designed & Developed by Vishali, Suraj, and Vaanavee

The Problem with Manual Tracking

- **Human Error in Counting:** Manual logging leads to input mistakes and incorrect tallying of items picked from shelves.
- **Inventory Mismatches:** Discrepancies between physical shelf stock and database records go unnoticed for long periods.
- **Delayed Discrepancy Detection:** Stock audits only reveal missing or misplaced items during periodic reviews, rather than in real-time.
- **Lack of Independent Verification:** Traditional systems record what a worker **claims** to have taken, without an independent, automated count check.

The Solution: RFID + AI Verification

- **Automated AI Cross-Check:** Every manual scan and RFID transaction is immediately cross-checked against a real-time computer vision count.
- **RT-DETR Object Detection:** A top-mounted AI camera running a fine-tuned RT-DETR model counts items in an enclosed verification tray.
- **Instant Alert Flagging:** Mismatches between the manual scan and the AI count raise a real-time alert, preventing incorrect data entries.
- **Complete Audit Trail:** Creates a fully verifiable, time-stamped log of who entered, what they picked, and what was verified by AI.

End-to-End Workflow

1. Labour Entry

Worker scans ID tag at entry door, logging entry time.

2. Rack ID

Worker scans rack QR code to view stored product catalogue.

3. Pick & Scan

Worker picks products and manually scans their RFID tags.

4. AI Verification

Worker places items on tray. Camera counts items using AI.

5. Data Logged

Matches are saved. Mismatches block auto-store and raise alerts.

System Components & Tech Stack

- **Frontend Dashboard:** Built with React, Vite, and Tailwind CSS. Provides live monitoring, analytics, and settings control.
- **Backend Services:** Node.js (for authentication, RFID tracking, and transactions) + Python FastAPI (for inventory logs, analytics, and RT-DETRv2 models).
- **Computer Vision & AI:** RT-DETRv2 neural network (runs on CPU/GPU via Hugging Face/PyTorch) for live object detection and ByteTrack for tracking.
- **Firmware & Hardware:** ESP32 microcontrollers deployed at entrance units and racks with RC522 RFID scanners and OLED displays.

Thank You!

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